

MISINFORMATION DETECTOR

for Immigration News in Spanish Digital Media

Milestone 2. Model Building and Evaluation

PROIII-06

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1. Introduction

Misinformation about migration in digital media represents a significant challenge for public understanding and policy, particularly in Spain. Social media platforms, online news outlets, and public communication channels can rapidly propagate false or misleading content, impacting political, social, and health-related discourse.

This project addresses the detection of misinformation by proposing an end-to-end data processing pipeline capable of ingesting content, classifying claims, retrieving supporting evidence, and providing explainable feedback through a chatbot interface. The focus is on Spanish political news, particularly migration-related content. The primary objective is to detect misinformation early, reducing the spread of them and equip users with tools to verify and understand false claims. This last point is important because free misinformation detection is not accessible for everyone.

2. Data Preparation

2.1 Dataset Construction

A custom dataset was created by scraping news from multiple verified Spanish sources (Cadena SER (n.d.), El País (n.d.), Agencia EFE (n.d.), Maldita Migración (n.d.)). All sources were unified into a common schema of eight columns: `titulo`, `etiqueta`, `texto`, `tema`, `url`, `fecha_publicacion`, `medio`, and `fecha_extraccion`. The initial real-only dataset contained news items labelled VERDADERO, FALSO, or CONTEXTO. Eleven news items categorized as ALERTA were reclassified as VERDADERO, as they were verified alerts.

We generated a synthetic dataset of manipulated news items from the real ones. The synthetic schema comprises seven columns: `titulo`, `etiqueta`, `texto`, `tema`, `tecnic`, `modificacion`, and `fecha_publicacion`. Synthetic items were labelled as either FALSO or CONTEXTO and then integrated with the real news.

2.2 Analysis of Label Distribution

Figure 1 shows a bar chart comparing the real-only distribution against the final combined dataset. The real dataset alone was highly imbalanced: VERDADERO articles dominated, while FALSO and CONTEXTO were scarce. To make model training feasible, we generated synthetic false or misleading variants from true articles using AI. The literature stresses that high-quality fake-news datasets should be representative, reliable, and diverse, and that labels, annotation rules and dataset biases directly affect downstream model performance. It also highlights that balanced class distributions usually reduce majority-class bias, but synthetic data should complement real distributions rather than hide their limitations (Kuntur, S.). The final combined dataset reduces the original 11:1 ratio to approximately 0.86:1, achieving a practically balanced corpus.

2.3 Cleaning and Transformation Pipeline

The preparation pipeline applied the following transformations to the news dataset:

- Null removal: Rows with a null title or text were eliminated using `dropna()`.
- Length filtering: Entries with a title shorter than 5 characters or a text shorter than 20 characters were discarded.
- Normalisation: Whitespace was stripped and UTF-8 encoding was standardised.
- Stratified split: The data was partitioned into 80% training, 10% validation, and 10% test sets using `train_test_split` with `stratify=label`, thereby preserving the class proportions in each partition.

Text length analysis of the news revealed a mean of approximately 1200 characters per item (title+text), with the 90th percentile at around 3000 characters, equivalent to 450-500 XLM-RoBERTa tokens. Concatenating the title and text into a single model input maximises the information available for classification. This concatenated input remains within the 512-token limit of XLM-RoBERTa for the 85th percentile of examples.

2.4 Data Understanding and Exploratory Data Analysis

To fully comprehend the nature of the misinformation in our statistics dataset, we conducted an Exploratory Data Analysis. We analyzed the main thematic pillars that drive migration-related news. We focused on three main themes for this analysis, using data from Instituto Nacional de Estadística. (n.d.). *INE*. and Ministerio de Inclusión, Seguridad Social y Migraciones. (n.d.).

1. Arrivals

Fake news often exaggerates migrant arrivals and their origins. Our 2024 choropleth map (*Figure 2*) clarifies the real situation, showing strong but defined migratory flows mainly from Latin America (specifically Colombia and Venezuela) and North Africa (Morocco).

2. Criminality

Disinformation frequently links migration directly to crime. Yet, in 2024, Spanish nationals committed the vast majority of total crimes (71.2%) (*Figure 3*) and highly sensitive offenses like sexual crimes, over 3,000 cases (*Figure 4*). While crimes by some foreign nationalities have risen since 2020 (*Figure 5*), fake news manipulates this data by ignoring broader contexts, like general population growth.

3. Economy

Fake articles claim migrants drain public resources or avoid working. Conversely, data proves that foreign groups have consistently higher employment rates than Spaniards (*Figure 6*), and the number of active foreign workers has grown significantly over the last decade (*Figure 7*). Although non-EU foreigners face higher unemployment (*Figure 8*), disinformation ignores that this rate has dropped dramatically since 2016.

This analysis confirms that misinformation deliberately manipulates real statistics across these three themes. This directly justifies our use of advanced language models like BETO or XLM-RoBERTa, as simple keyword approaches cannot capture the deeper semantic context and alarmist tone required to detect fake news.

3. Task description

The task is a classification problem. Given a news item about migration, the system must classify it into one of the categories: VERDADERO, FALSO, CONTEXTO. The input is the text of the news item; the output is a label, a confidence score, and an analysis.

The model is designed to work as part of a larger system:

The model analyses the text and returns a label (VERDADERO, FALSO o CONTEXTO), a probability for each one, and an analysis in which you can check the origin of the main source that published it, the real data, how it has been spread...

The RAG system searches for similar verified news items as evidence.

The chatbot's LLM uses the model's output and the RAG evidence to generate an explained response in natural language.

The model's sole function is to provide the classification and confidence. It does not explain why a news item is false, that is done by the LLM with RAG evidence.

3.1 Materialising minable view

The problem is formulated as a supervised text classification task.

- Input (X): Statement of a controversial new, a person's social opinion, ...
- Output (Y): A categorical label indicating whether the news is VERDADERO, FALSO, or CONTEXTO.
- Sample unit: One individual news item.
- Dataset size: The final dataset consists of 5,740 news items.

In terms of feature representation, a baseline approach using TF-IDF vectors was considered but discarded due to its inability to capture semantic nuance. The final models use contextual embeddings from fine-tuned Transformer encoders (BETO and XLM-RoBERTa), which capture deeper semantic and contextual information in Spanish text, improving classification performance.

4. Model prototype and evaluation

4.1 Model Selection and Architecture

We fine-tune two Transformer-based models: BETO (dccuchile/bert-base-spanish-wwm-cased), pretrained exclusively on Spanish, and XLM-RoBERTa (xlm-roberta-base), a multilingual model. This allows us to compare domain specialisation versus generalisation capacity. Traditional approaches such as TF-IDF with logistic regression or SVM were discarded because they fail to capture the semantic nuance required to distinguish CONTEXTO from VERDADERO a distinction that depends on subtle framing and context rather than keyword frequency. Both models share the same architecture: a pretrained Transformer encoder, a fully connected classification head, and a softmax output over 3 classes. The input is the text, tokenized and truncated to 512 tokens.

4.2 Training Configuration

Both models are trained with AdamW, cosine LR scheduler with 10% warmup, early stopping (patience = 3), and FP16 mixed precision. Key hyperparameters are summarised below (table 1):

Hyperparameter	BETO	XLM-RoBERTa
Learning rate	2e-5	1.5e-5
Batch size (per GPU)	16	32
Max epochs	6	6
Best model metric	F1_falso	F1_macro

Table 1: Hyperparameters used for each model

The learning rate and batch size differences are justified by model size and available hardware respectively. To handle class imbalance, both models apply inverse-frequency class weights and label smoothing ($\alpha = 0.1$), which reduces overconfidence and improves probability calibration. A confidence threshold of 0.60 flags low-confidence predictions as INCIERTO rather than forcing an incorrect classification. Different evaluation metrics were used during training. BETO uses F1 for the FALSO class, while RoBERTa uses macro F1. This results in different optimization behaviours and is treated as part of the experimental comparison.

4.3 Integration and Experimental Strategy

The classifier is one component of a larger pipeline: it produces a label, confidence score, and a `prompt_hint` that is passed to an LLM, which combines this verdict with documents retrieved by a RAG system to generate a full forensic explanation for the user. Both models are fully trained and evaluated under identical conditions; XLM-RoBERTa is currently deployed in the chatbot while final model selection will be determined by the evaluation results in Section 5.

4.4 Evaluation Metric Justification

Given the multiclass nature of the task and the class imbalance present in the dataset, accuracy alone is insufficient as an evaluation metric. We use macro F1-scores as the primary metric, which treats all classes equally regardless of their frequency. We also report per-class precision, recall and F1, as well as weighted F1 and overall accuracy. This combination allows us to assess both global performance and the model's behaviour on each individual class, particularly the minority class CONTEXTO.

4.5 Results

Both models were evaluated on the same held-out test set. Results are summarised below (table 2):

Metric	BETO	XLM-RoBERTa
Accuracy	0.838	0.846
F1 Macro	0.749	0.782
F1 Weighted	0.824	0.843
F1 VERDADERO	0.991	0.994
F1 CONTEXTO	0.455	0.554
F1 FALSO	0.801	0.798

Table 2: Comparing important metrics for both models

Per-class results – BETO (table 3):

Class	Precision	Recall	F1	Support
VERDADERO	0.992	0.989	0.991	264
CONTEXTO	0.633	0.355	0.455	107
FALSO	0.724	0.896	0.801	202

Table 3: Results of different metrics per-class in BETO model

Per-class results – XLM-RoBERTa (table 4):

Class	Precision	Recall	F1	Support
VERDADERO	1.000	0.989	0.994	264
CONTEXTO	0.614	0.505	0.554	107
FALSO	0.759	0.842	0.798	202

Table 4: Results of different metrics per-class in XLM- RoBERTa model

4.6 Training Behaviour

BETO converged in 4 epochs (early stopping triggered), with the best checkpoint selected from epoch 1 (F1_falso = 0.785) (Table 5).

XLM-RoBERTa ran all 6 epochs, reaching its best macro F1 of 0.783 at epoch 4 (Table 6).

Both models show a steady decrease in training loss, with validation loss stabilising after epoch 2, indicating no significant overfitting.

4.7 Analysis

Both models perform strongly on VERDADERO ($F1 > 0.99$) and achieve solid results on FALSO ($F1 \approx 0.80$), the most critical class for misinformation detection. The most challenging class is CONTEXTO (BETO: 0.455, XLM-RoBERTa: 0.554), which is expected given its smaller size, its linguistic similarity to VERDADERO, and the inherent ambiguity of the class itself. XLM-RoBERTa outperforms BETO on CONTEXTO by +0.10 F1 and achieves higher macro F1 overall (0.782 vs 0.749), suggesting its multilingual pretraining generalises better to subtle semantic distortions. Based on these results, XLM-RoBERTa is selected for deployment in the final pipeline.

4.8 Deployment mockup

A functional Telegram prototype called FakeNewsbot was developed to simulate the real use of the system. The system integrates three components connected through the bot, the model and the RAG (Figure 9).

The user sends a claim or news sentence through the chat interface, and the bot returns: the predicted class generated by the classifier, the confidence scores for each class, and a short forensic explanation generated with the RAG + LLM module, supported by retrieved verified evidence. This prototype shows how the complete pipeline works in practice and demonstrates that the system can assist users not only by detecting potentially false migration-related claims, but also by providing an interpretable justification for the decision. In the Figure 10 and Figure 11 we can see a real interaction with the prototype, where we sent two headlines that were classified as false and true respectively, and the answer including classification output, confidence distribution and evidence-based explanation. Furthermore, a final section was included that gives other related news and their links.

4.9 Limitations

Both models show low confidence on short headline-only inputs (all predictions below the 0.60 threshold in the inference test), confirming that the models require full article text to produce reliable predictions. This is consistent with the training setup, where both title and body were always provided as input.

5. Discussion

5.1 Use of technology

Python was the primary language throughout the project. For model training, Hugging Face Transformers with PyTorch was used to fine-tune both BETO and XLM-RoBERTa, trained on Google Colab and Kaggle respectively using NVIDIA T4 GPUs. Practical challenges such as VRAM management, FP16 mixed precision, and checkpoint handling were solved autonomously.

For the retrieval system, ChromaDB was used as the vector database, with embeddings generated via the Gemini Embedding API. The Gemini API (gemini-2.5-flash) served as the LLM for generating forensic explanations. The full pipeline was deployed through two interfaces: a local web app built with Gradio and a functional Telegram bot, both integrating the classifier, RAG system, and LLM in real time.

5.2 Seminar and AI Use

The project applies key concepts from the course seminars across all stages. The problem is formulated as a supervised classification task (Unit 3) with clearly defined input and output variables and a well-specified minable view. Model selection followed the criteria discussed in the seminars: alternative approaches such as TF-IDF with Logistic Regression were considered and discarded in favour of Transformer-based models, justifying why they are more appropriate for capturing semantic nuance in Spanish misinformation detection. Evaluation was performed using macro F1-score, chosen specifically because of class imbalance — a metric selection criterion covered in Unit 4. The deployment prototype (Unit 5) demonstrates a first estimate of the system's practical value through a functional Telegram bot and Gradio interface.

AI tools were used in two controlled ways: to generate synthetic data addressing class imbalance, and as part of the RAG pipeline to produce evidence-based explanations. Both uses are justified and documented throughout the report.

6. References

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Cañete, J., Chaperón, G., & Pérez, J. (2020, February 7). *BETO: la red neuronal que podría ser utilizada durante el próximo proceso constituyente*. Instituto Milenio Fundamentos de los Datos. <https://imfd.cl/notwg2025-beto-la-red-neuronal-que-podria-ser-utilizada-durante-el-proximo-proceso-constituyente/>

7. Appendices

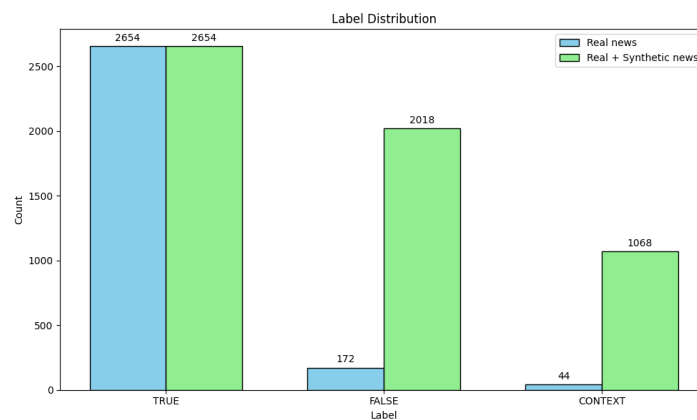


Figure 1: Label distribution of real news versus combined real and synthetic news

Migraciones por país en 2024

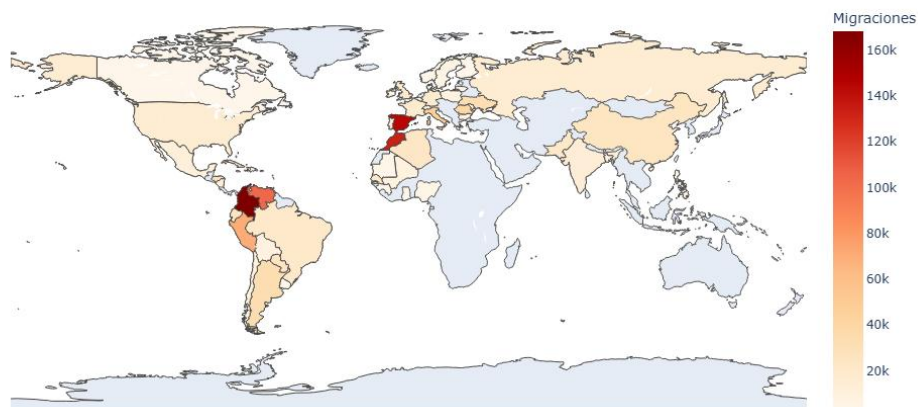


Figure 2: Geographic distribution of migratory flows to Spain by country of origin (2024).

Distribución de top 10 nacionalidades en 2024 para A Delitos (excluyendo Total)

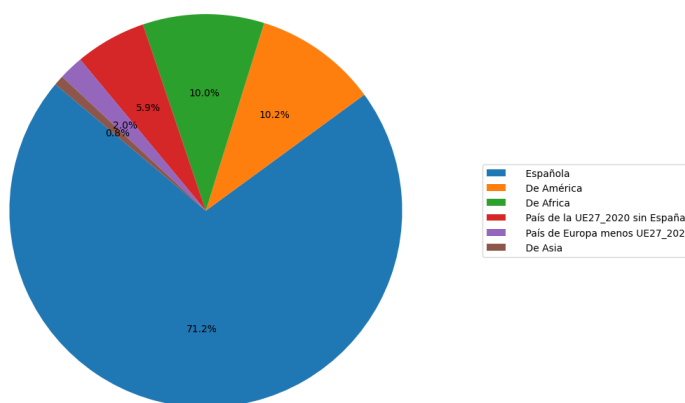


Figure 3: Pie chart showing the percentage of total crimes by nationality in Spain (2024).

Top 10 nacionalidades en 2024 para 8 Contra la libertad e indemnidad sexuales (excluyendo Total)

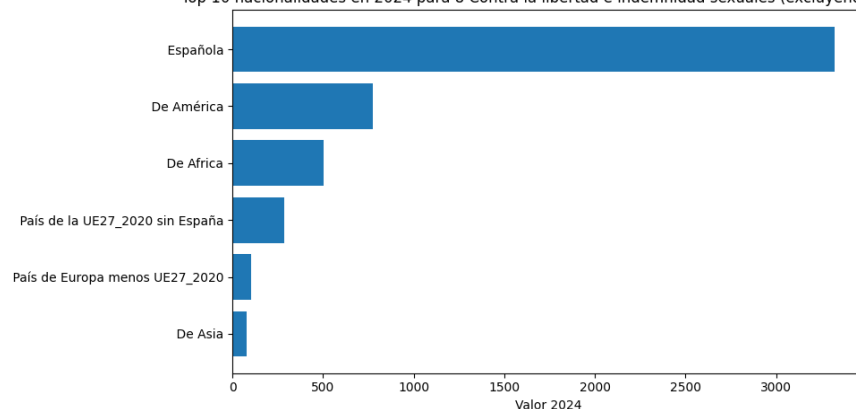


Figure 4: Bar graph showing the number of sexual crimes by nationality in Spain (2024).

Evolución por año (top 6 nacionalidades, excluyendo Total y Española) en A Delitos

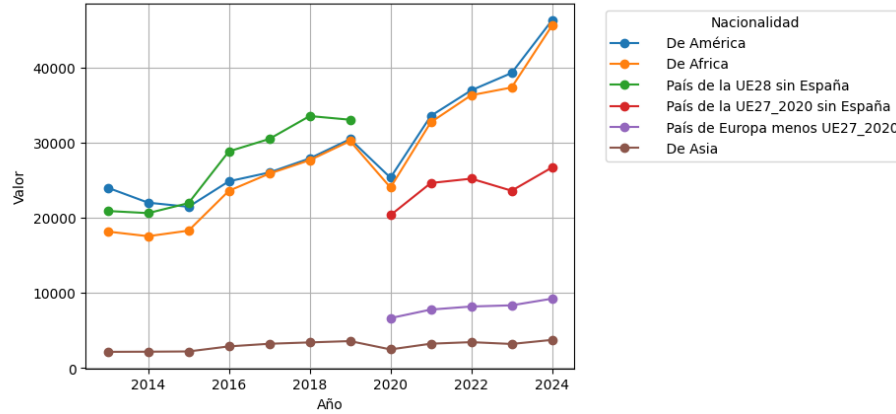


Figure 5: Line graph showing the evolution of crimes committed by the main foreign nationalities in Spain (2013-2024).

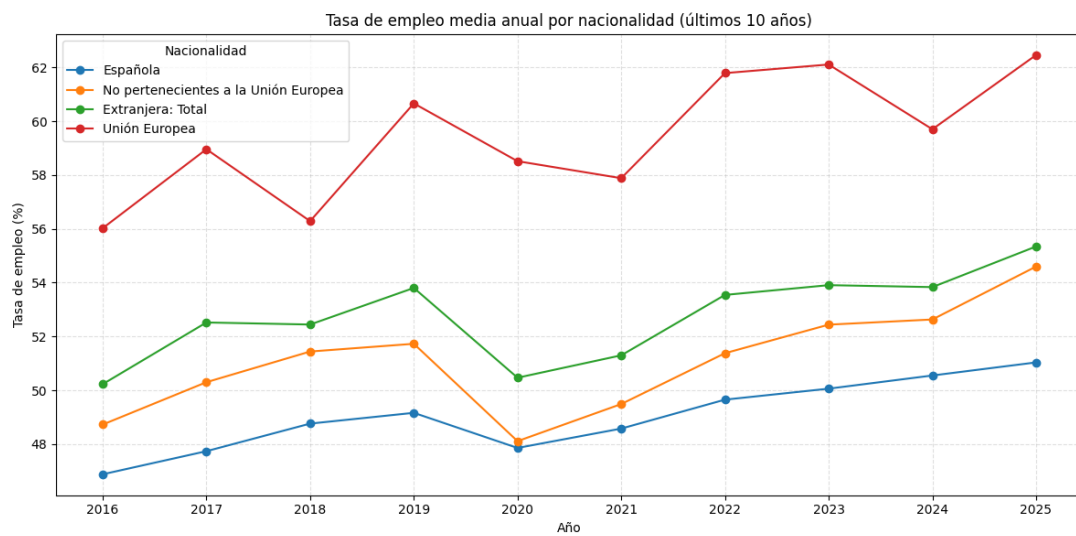


Figure 6: Line chart showing the average annual employment rate by nationality in Spain (2016-2025).

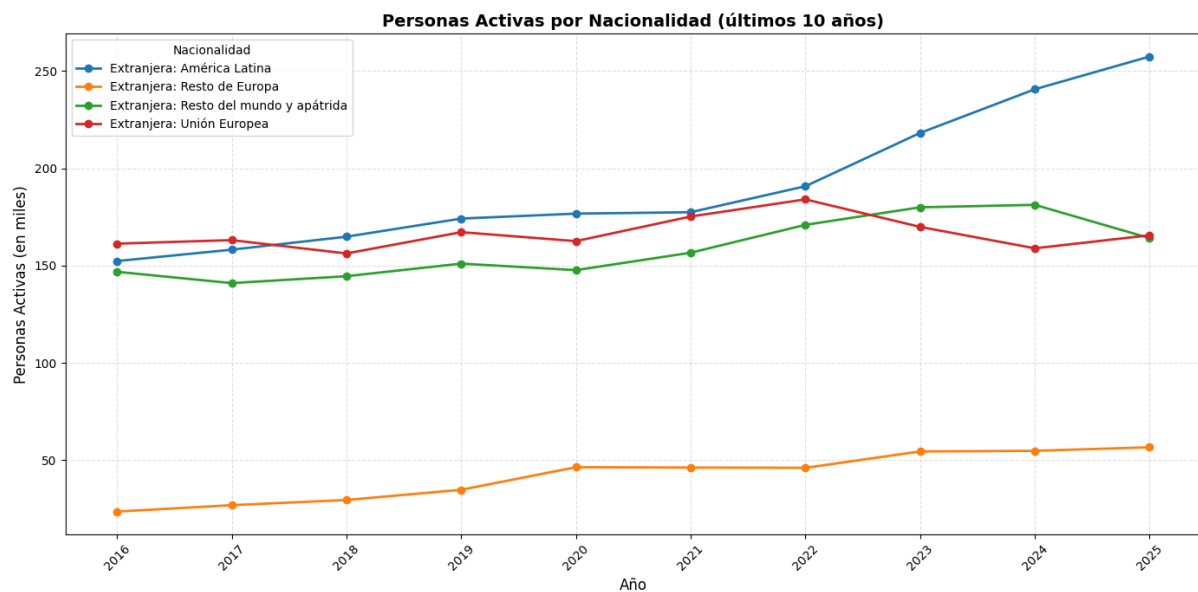


Figure 7: Line chart showing the evolution of active foreign workers in Spain by region of origin (2016-2025).

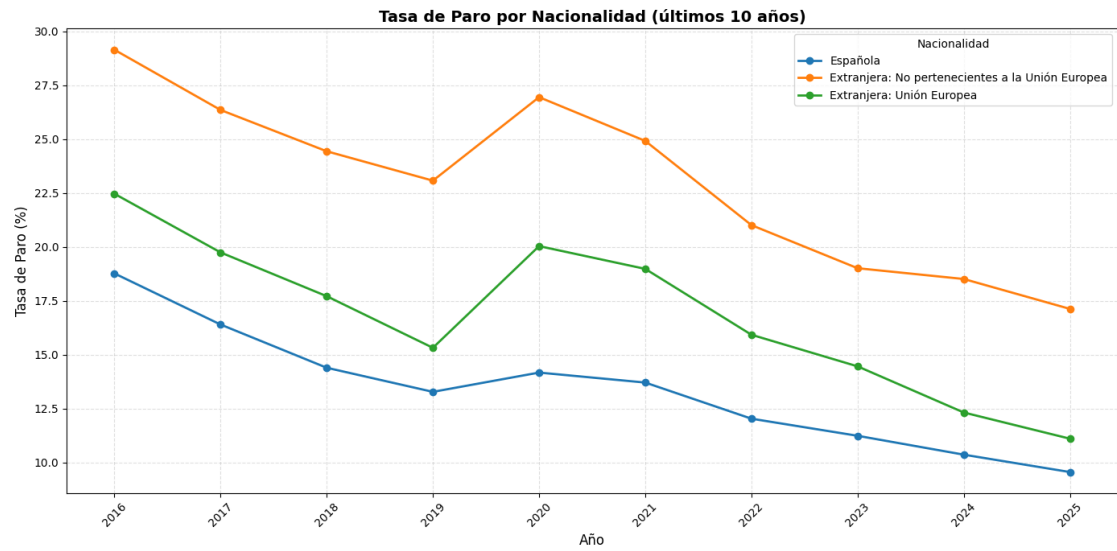


Figure 8: Line chart showing the unemployment rate by nationality in Spain (2016–2025).

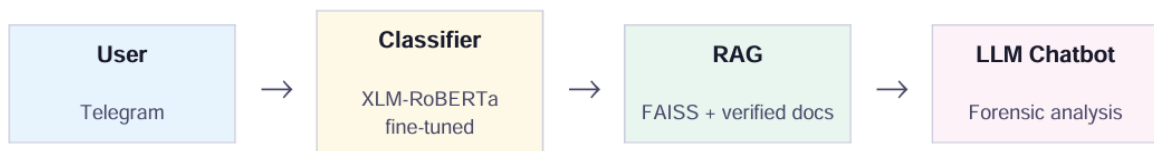


Figure 9: Full pipeline architecture

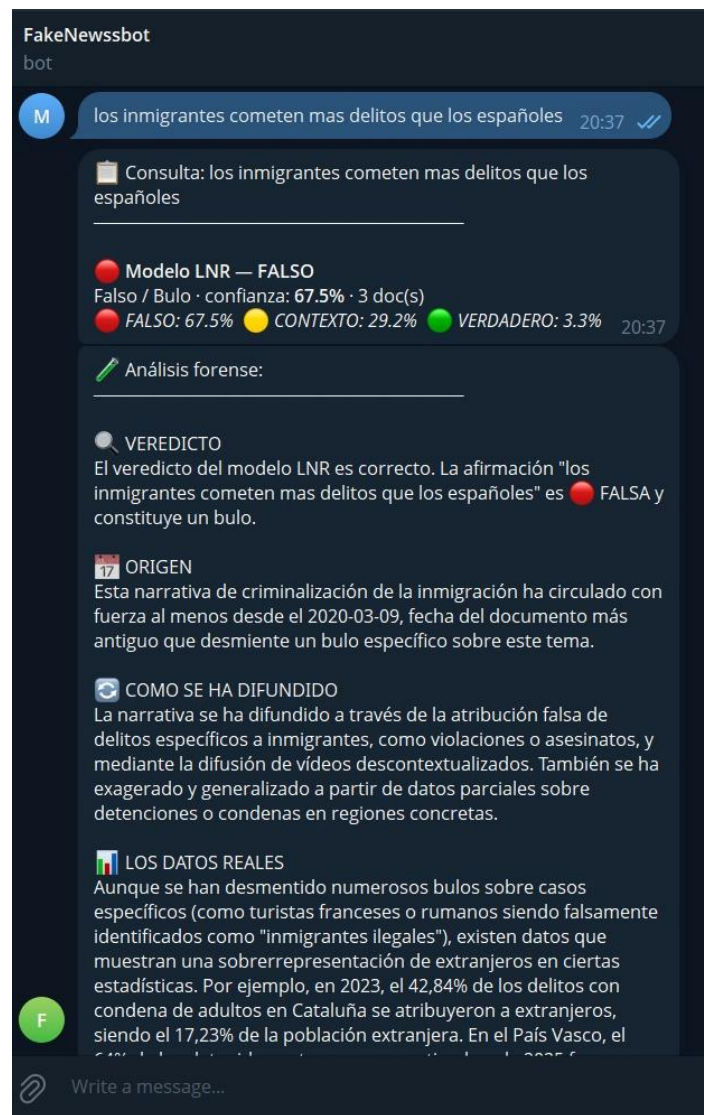


Figure 10: Example of the answer of the bot to a news headline, classified as false

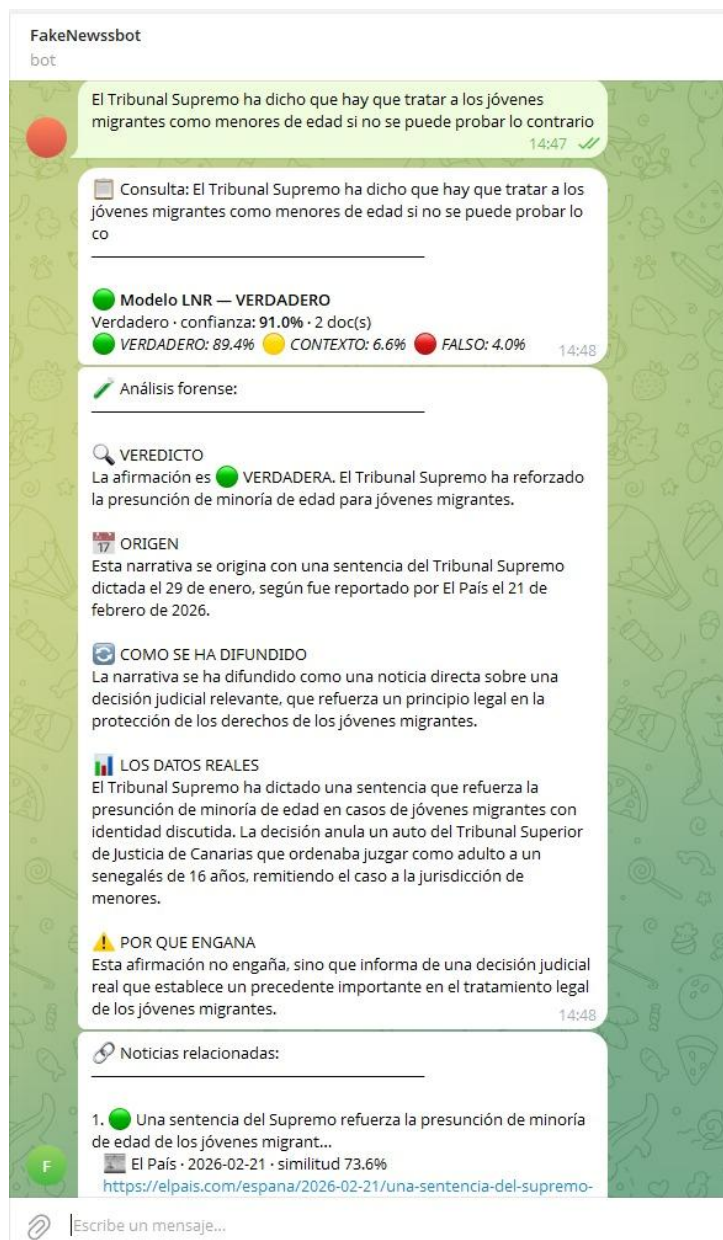


Figure 11: Example of the answer of the bot to a news headline, classified as true

8. Table index

Table 1: Hyperparameters used for each model

Table 2: Comparing important metrics for both models

Table 3: Results of different metrics per-class in BETO model

Table 4: Results of different metrics per-class in XLM- RoBERTa model

Table 5: Results of training of BETO model

Epoch	Training Loss	Validation Loss	Accuracy	F1 Macro	F1 Weighted	F1 Falso
1	0.672436	0.662023	0.825480	0.716814	0.805232	0.785249
2	0.619711	0.622900	0.808028	0.762769	0.810799	0.666667
3	0.547921	0.634254	0.834206	0.782159	0.836961	0.758974
4	0.443641	0.735547	0.837696	0.779897	0.838085	0.773399

Table 6: Results of training of Roberta Model

Epoch	Training Loss	Validation Loss	Accuracy	F1 Macro	F1 Weighted	F1 Falso
1	0.978531	0.741924	0.794066	0.677988	0.766136	0.753304
2	0.691416	0.640365	0.818499	0.771065	0.823403	0.714681
3	0.606289	0.642274	0.792321	0.742262	0.796972	0.658824
4	0.555036	0.645879	0.849913	0.783438	0.844633	0.800926
5	0.539183	0.659617	0.834206	0.776359	0.834578	0.767327
6	0.517412	0.676319	0.835951	0.776276	0.835409	0.772616